

## Effect of blueberry ingestion on natural killer cell counts, oxidative stress, and inflammation prior to and after 2.5 h of running.

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Blueberries are rich in antioxidants known as anthocyanins, which may exhibit significant health benefits. Strenuous exercise is known to acutely generate oxidative stress and an inflammatory state, and serves as an on-demand model to test antioxidant and anti-inflammatory compounds. The purpose of this study was to examine whether 250 g of blueberries per day for 6 weeks and 375 g given 1 h prior to 2.5 h of running at ~72% maximal oxygen consumption counters oxidative stress, inflammation, and immune changes. Twenty-five well-trained subjects were recruited and randomized into blueberry (BB) (N = 13) or control (CON) (N = 12) groups. Blood, muscle, and urine samples were obtained pre-exercise and immediately postexercise, and blood and urine 1 h postexercise. Blood was examined for F<sub>2</sub>-isoprostanes for oxidative stress, cortisol, cytokines, homocysteine, leukocytes, T-cell function, natural killer (NK), and lymphocyte cell counts for inflammation and immune system activation, and ferric reducing ability of plasma for antioxidant capacity. Muscle biopsies were examined for glycogen and NFκB expression to evaluate stress and inflammation. Urine was tested for modification of DNA (8-OHdG) and RNA (5-OHmU) as markers of nucleic acid oxidation. A 2 (treatment) × 3 (time) repeated measures ANOVA was used for statistical analysis. Increases in F<sub>2</sub>-isoprostanes and 5-OHmU were significantly less in BB and plasma IL-10 and NK cell counts were significantly greater in BB vs. CON. Changes in all other markers did not differ. This study indicates that daily blueberry consumption for 6 weeks increases NK cell counts, and acute ingestion reduces oxidative stress and increases anti-inflammatory cytokines.